Viral Content Agent — Product Requirements Document

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1. Problem Statement

Content creators, agencies, and researchers spend hours manually scrolling platforms to find viral content — and still walk away with vague intuitions rather than actionable patterns. There is no systematic way to (a) surface the right viral clips across platforms, and (b) decompose *why* they worked in a reproducible format.

The gap: existing tools (e.g. Exploding Topics, SparkToro) identify trends at the topic level. None reverse-engineer the structural mechanics of individual viral clips and store those mechanics in a reusable library.

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2. Vision

An AI agent that continuously monitors TikTok, Instagram Reels, YouTube Shorts, Twitter/X, and Reddit for viral content, extracts the structural elements that made each piece work, and surfaces those elements as a reusable pattern library — enabling creators and strategists to replicate performance, not just observe it.

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3. Goals

Goal	Success Metric
Surface viral content faster than manual search	Agent delivers results in < 2 min; manual baseline is ~45 min
Identify replicable patterns, not just content	⩾ 80% of extractions have ⩾ 4 populated element fields
Support cross-platform coverage	All 5 platforms active at MVP launch
Enable both reactive and proactive workflows	Both scheduled reports and on-demand search functional at launch

Goal	Success Metric
Serve creators, agencies, and researchers	Auth supports personal + team workspace

Non-Goals (v1)

- Video editing or content generation (downstream tool, not this agent)
- Paid ad analysis
- Sentiment analysis of comments at scale
- Audio/speech transcription (v2)

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4. Users

Primary Personas

Solo Creator
Runs their own content on 1-2 platforms. Wants a weekly digest of what's popping in their niche so they can reverse-engineer hooks and formats without spending hours on research.

Content Agency
Manages 5-20 client accounts across niches. Needs to run targeted searches per client, build shareable reports, and maintain a growing library of proven patterns per vertical.

Competitor Researcher / Brand Strategist
Tracks competitor accounts and industry hashtags. Wants structured data exports, not just video links — what format, what hook type, what emotional trigger, what CTA.

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5. User Stories

Search & Discovery

- As a creator, I want to search by niche/keyword and get the top viral clips from the last 7/30/90 days so I can see what's performing right now.
- As an agency, I want to run a search scoped to a competitor's account or hashtag so I can track their content strategy.
- As a user, I want to filter results by platform, virality score, content format, and date range so I can narrow to exactly what I need.

Element Extraction

- As a creator, I want each viral clip broken down into its structural elements (hook, pacing, format, emotion, CTA) so I know *what to replicate*, not just what to watch.
- As a strategist, I want the agent to tag patterns across clips so I can see "the 3-second question hook" appears in 12 of the top 20 videos this week.
- As an agency, I want to save and annotate extracted elements so I can build a client-specific pattern library over time.

Scheduled Reports

- As a user, I want to configure weekly/daily digests per niche or account list, delivered to email or a dashboard, so I stay informed without manual effort.
- As a team lead, I want to share a scheduled report with a client so they receive a branded PDF/link of this week's top viral content in their space.

Library & Reuse

- As a creator, I want a saved pattern library organized by element type (hook styles, format types, CTA patterns) so I can pull from it when planning new content.
- As a user, I want to search my library by element tag so I can find "all clips that used a 'before/after' format" across any niche.

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6. Core Features

6.1 Platform Connectors

Platform	Data Source	Key Signals
TikTok	TikTok Research API / scrape fallback	Views, shares, comments, watch-through rate, shares-to-views ratio
Instagram Reels	Meta Graph API (Business)	Reach, saves, shares, plays
YouTube Shorts	YouTube Data API v3	Views, likes, comments, retention signal
Twitter/X	X API v2 (Basic tier)	Views, retweets, bookmarks, reply ratio

Platform	Data Source	Key Signals
Reddit	Reddit API (OAuth)	Upvotes, upvote ratio, cross-posts, comment velocity

Virality Score Formula (v1)
A weighted composite per platform. Example for TikTok:

```
virality_score = (
  0.35 * normalized_shares +
  0.25 * normalized_watch_through +
  0.20 * normalized_views_velocity +
  0.10 * normalized_comments +
  0.10 * normalized_saves
)
```

Weights are platform-specific and tunable per user in settings.

6.2 Search Agent (On-Demand)

Inputs

- Query: keyword, hashtag, account handle, or topic
- Platform(s): multi-select
- Timeframe: 24h / 7d / 30d / 90d
- Virality threshold: minimum score (default: top 5%)
- Content format filter: video, carousel, text post, thread

Agent Loop (LangGraph)

```
START
  → PlatformFetcher (parallel, per platform)
  → ViralityRanker (score + filter)
  → ContentSampler (top N clips selected)
  → ElementExtractor (LLM analysis per clip)
  → PatternAggregator (cross-clip pattern detection)
  → ResultFormatter (structured output)
END
```

Output per search

- Ranked list of clips with virality score

- Element breakdown per clip (see §6.3)
- Cross-clip pattern summary ("3 of top 5 used a 'controversy hook'")
- Save to library option

6.3 Element Extractor

The core intelligence layer. For each viral clip, the agent produces a structured `ContentDNA` object:

```
json
```

```
{
  "clip_id": "uuid",
  "source_url": "https://...",
  "platform": "tiktok",
  "virality_score": 0.91,
  "posted_at": "2026-03-22T14:30:00Z",
  "niche": "fitness",

  "hook": {
    "type": "question | controversy | bold_claim | pattern_interrupt | before_after
    "duration_seconds": 3,
    "text_verbatim": "Nobody talks about why...",
    "delivery": "text_overlay | spoken | visual_action"
  },

  "format": {
    "type": "talking_head | voiceover_b_roll | text_post | carousel | duet | stitch
    "length_seconds": 47,
    "pacing": "slow | medium | fast | variable",
    "cut_frequency": "low | medium | high",
    "caption_style": "none | minimal | heavy_text | subtitles"
  },

  "emotion": {
    "primary": "curiosity | anger | inspiration | humor | fear | nostalgia | awe | r
    "secondary": "string | null",
    "intensity": 1-5
  },

  "structure": {
    "act_1_hook": "What happens in seconds 0-3",
    "act_2_body": "Core content pattern summary",
    "act_3_close": "How it ends / CTA",
    "loop_factor": true | false
  },

  "cta": {
    "type": "comment | follow | share | link_in_bio | none | implicit",
    "placement": "verbal | on_screen | caption | none",
    "text_verbatim": "Drop a 🔥 if this helped"
  },

  "replication_notes": "Free text: what specifically to steal from this clip",
```

```
  "pattern_tags": ["3-second-question-hook", "before-after", "high-cut-frequency"]
}
```

Extraction method: LLM (Claude Sonnet) is passed the clip URL + metadata + transcript (if available). Prompt instructs the model to fill the `ContentDNA` schema. For video-only content, frame samples are extracted and passed as images.

6.4 Scheduled Reports

Configuration per report

- Name, niche/keywords, accounts to monitor
- Platform(s)
- Frequency: daily / weekly / custom cron
- Delivery: dashboard notification + email digest + optional PDF export
- Top N results (default: 10)
- Include: pattern summary, element breakdowns, links

Report structure

1. Executive summary: this week's top patterns
2. Top clips table (ranked by virality score)
3. Per-clip ContentDNA breakdown (expandable)
4. Pattern delta: new patterns vs. last report
5. Recommended replications: 3 content brief suggestions generated by LLM

6.5 Pattern Library

A persistent, searchable store of saved `ContentDNA` objects.

Organization

- Auto-tagged by element fields (hook type, format, emotion, platform)
- User-created collections (e.g. "Q2 Fitness Campaign", "Client: Acme")
- Niche folders

Search

- Full-text search across all fields

- Filter by any `ContentDNA` field
- Sort by virality score, recency, times-reused

Team features (v1.5+)

- Shared workspaces
- Annotation / notes per saved clip
- Export collection as CSV or PDF brief

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7. Data Model (High-Level)

```
User
  └── Workspaces (personal / team)
        ├── SearchQueries (on-demand, saved history)
        ├── ScheduledReports (config + delivery history)
        ├── Collections (named pattern libraries)
        |     └── ContentDNA[] (saved clips + extractions)
        └── Settings (virality weights, platform tokens)

Platform
  └── RawClip (url, metadata, cached signals)
        └── ContentDNA (extracted elements, linked 1:1)
              └── PatternTag[] (many-to-many, cross-clip)
```

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8. Tech Stack Recommendation

Layer	Choice	Rationale
Agent orchestration	LangGraph	Stateful agent loops, parallel node execution, built-in retry
LLM	Claude Sonnet (via API)	Vision + text, structured output, your existing familiarity
Platform data	Per-platform API + Apify fallback	Apify covers scrape-only platforms (TikTok public, IG)

Layer	Choice	Rationale
Vector DB	Qdrant (self-hosted) or Pinecone	Pattern similarity search across ContentDNA embeddings
Backend	FastAPI (Python)	Async, plays well with LangGraph + httpx
Database	Postgres (Neon serverless)	Low cost, matches your existing Neon familiarity
Task queue	Celery + Redis	Scheduled report jobs
Frontend	Next.js + Tailwind	Dashboard + search UI
Auth	Clerk or Supabase Auth	Fast to ship, supports teams
Deployment	Railway or Render	Low ops overhead for solo/small team

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9. Phased Roadmap

Phase 1 — MVP (Weeks 1-6)

Goal: On-demand search + element extraction working for 2 platforms

- [ ] Platform connectors: TikTok + YouTube Shorts
- [ ] Virality scorer (views velocity + shares)
- [ ] LangGraph agent loop: fetch → rank → extract
- [ ] ContentDNA schema + Claude extraction prompt
- [ ] Basic web UI: search input, results list, ContentDNA card per clip
- [ ] Manual save to library

Exit criteria: User can search a niche, get top 10 viral clips, see a ContentDNA breakdown for each, save to library.

Phase 2 — Full Platform Coverage (Weeks 7-10)

Goal: All 5 platforms live + scheduled reports

- [ ] Instagram Reels, Twitter/X, Reddit connectors
- [ ] Scheduled report builder UI
- [ ] Email digest delivery
- [ ] Pattern aggregator: cross-clip tag detection
- [ ] Library search + filter by element fields

- [ ] PDF export for reports

Exit criteria: User can configure a weekly digest for any niche across all 5 platforms.

Phase 3 — Intelligence Layer (Weeks 11-16)

Goal: The library gets smarter over time

- [ ] Vector embeddings for ContentDNA objects (Qdrant)
- [ ] Similarity search: "find clips like this one"
- [ ] Pattern delta in reports ("this hook type is trending up +40% this week")
- [ ] LLM-generated content brief from selected patterns
- [ ] Team workspaces + shared collections
- [ ] Competitor account monitoring

Exit criteria: User can select 3 saved patterns and generate a ready-to-shoot content brief.

Phase 4 — Growth Layer (Weeks 17+)

- Audio transcription + hook text extraction (Whisper)
- Chrome extension: extract ContentDNA from any video while browsing
- API access for agencies / developers
- Benchmark mode: score your own clips against viral patterns
- White-label PDF reports for agency clients

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10. Key Risks & Mitigations

Risk	Likelihood	Mitigation
Platform API rate limits / access changes	High	Apify scrape fallback; abstract connector interface for swap-out
LLM extraction quality on short clips	Medium	Fine-tune prompt with structured examples; human review mode
Virality score gaming / outliers	Medium	Normalize per niche, not globally; flag statistical anomalies

Risk	Likelihood	Mitigation
Data freshness lag	Low-Medium	Cache TTL per platform; real-time mode for Twitter/X
Scraping ToS exposure	Medium	Prioritize official APIs; consult ToS per platform; add user auth tokens where available

11. Success Metrics

Metric	Target (3 months post-launch)
Searches run per active user / week	⩾ 5
ContentDNA extractions with ⩾ 4 fields populated	⩾ 85%
Scheduled reports configured per user	⩾ 2
Library items saved per user	⩾ 20
Report open rate (email)	⩾ 40%
D30 retention	⩾ 35%

12. Open Questions

1. **Monetization model**: Freemium (limited searches/month) vs. seat-based for teams?
2. **TikTok Research API access**: Requires academic/business approval — do we pursue or default to Apify from day one?
3. **ContentDNA quality bar**: Do we launch with a human review queue or trust LLM extraction fully from the start?
4. **Reddit scope**: Subreddits only, or also track individual viral posts in r/all?
5. **Clip caching**: Do we store video thumbnails/frames, or link-only to avoid storage costs?